

1. (a) For statement variables p , q and r , prove that

$$\left[(p \vee \sim q) \wedge (q \vee r) \right] \rightarrow (p \vee r)$$

is a tautology.

You may use either truth tables or logical equivalences. If you use logical equivalences, you should name the laws you use at each step.

(4 marks)

Truth table:

p	q	r	$(p \vee \sim q) \wedge (q \vee r) \rightarrow (p \vee r)$					
T	T	T	T	T	F	T	T	T
T	T	F	T	T	F	T	T	T
T	F	T	T	T	T	T	T	T
T	F	F	T	T	T	F	F	T
F	T	T	F	F	F	F	T	T
F	T	F	F	F	F	F	T	F
F	F	T	F	T	T	T	T	T
F	F	F	F	T	T	F	F	F


 All true
 ∴ Tautology

- (b) Find values of the statement variables a , b and c that show that the following argument is invalid. You do not need to show your working.

1. $a \rightarrow b$
 2. $\sim a \rightarrow c$
 3. $\sim b \rightarrow c$
- $\therefore b$

(2 marks)

b FALSE

a FALSE

c TRUE

$\rightarrow$ All premises T
Conclusion F

2. (a) Prove the following statement:

For all $a, b \in \mathbb{N}$, if a is a perfect square and b is not a perfect square then $a \times b$ is not a perfect square.

Recall that $\mathbb{N} = \{1, 2, 3, \dots\}$ represents the natural numbers, and a perfect square is a number of the form x^2 where x is an integer.

(4 marks)

Proof by contradiction:

Let $a, b \in \mathbb{N}$, $a = \text{PS}$, $b \neq \text{PS}$, $ab = \text{PS}$.

Then: In prime decomp. of a , ab ,
all exponents are even.

$\Rightarrow b = \frac{ab}{a}$ has all exponents even

$\Rightarrow b$ is a P.S.

CONTRADICTION

(b) Find a counterexample that disproves the following statement:

For all $x \in \mathbb{R}$, $\lfloor x^2 \rfloor = \lfloor x \rfloor^2$.

You do not need to show your working.

Recall that $\lfloor \cdot \rfloor$ denotes the floor function.

(2 marks)

$$x = 2.9 \Rightarrow \lfloor x^2 \rfloor = 8, \quad \lfloor x \rfloor = 2$$

$$\therefore \lfloor x \rfloor^2 = 4.$$

2. (c) Using the Euclidean algorithm, compute $\gcd(116, 88)$. Show your working.

(3 marks)

$$116 = 1 \cdot 88 + 28$$

$$88 = 3 \cdot 28 + 4$$

$$28 = 7 \cdot 4 + 0$$

$$\therefore \gcd = 4$$

- (d) Using your answer to part (c), compute $\text{lcm}(116, 88)$.

(2 marks)

$$\begin{aligned} \text{lcm} &= 116 \cdot 88 / \gcd \\ &= 116 \cdot 88 / 4 = 2552. \end{aligned}$$

3. Let $a_0, a_1, a_2, \dots$ be the sequence defined recursively by $a_0 = 0$, $a_1 = 1$, and

$$a_n = a_n = 3a_{n-1} - 2a_{n-2} \text{ for all integers } n \geq 2.$$

- (a) Compute the terms a_2 , a_3 and a_4 .

(3 marks)

$$a_2 = 3$$

$$a_3 = 7$$

$$a_4 = 15$$

- (b) Guess an explicit formula for this sequence, and prove that your formula is correct.

(Hint: If you have trouble guessing, you may find that computing more terms can help.)

(4 marks)

$$a_n = 2^n - 1$$

Proof:

$$\text{Base cases: } a_0 = 0 = 2^0 - 1 \checkmark$$

$$a_1 = 1 = 2^1 - 1 \checkmark$$

$$\text{Recurrence: } a_n = 3a_{n-1} - 2a_{n-2}$$

$$\Leftrightarrow 2^n - 1 = 3(2^{n-1} - 1) - 2(2^{n-2} - 1)$$

$$\Leftrightarrow 2^n - 1 = 3 \cdot 2^{n-1} - 3 - 2^{n-1} + 2$$

$$\Leftrightarrow 2^n - 1 = 2 \cdot 2^{n-1} - 1$$

$$\Leftrightarrow 2^n - 1 = 2^n - 1 \checkmark$$

Formula satisfies definition $\rightarrow$ correct.

4. (a) Write out the elements of $\{1, 2\} \cap \{1, 3\}$.

(1 mark)

$$\{1\}$$

- (b) Write out the elements of $\{1, 2\} \cup \{1, 3\}$.

(1 mark)

$$\{1, 2, 3\}$$

- (c) Write out the elements of $\{1, 2\} \times \{1, 3\}$.

(2 marks)

$$\{(1, 1), (1, 3), (2, 1), (2, 3)\}$$

- (d) Write out the elements of $\mathcal{P}(\{1, 2\})$.

(2 marks)

$$\{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

- (e) What is the cardinality of $\mathcal{P}(\mathcal{P}(\{1, 2\}))$? You should write your answer as a single integer. You do not need to show your working.

(2 marks)

$$2^4 = 16.$$

5. Consider the function $f : \mathbb{R} \rightarrow \mathbb{Z}$ defined by $f(x) = \lfloor x^2 \rfloor$.

Recall that $\lfloor \cdot \rfloor$ denotes the floor function.

(a) What is the image of the interval $[-1, 1]$? You do not need to show your working.

(2 marks)

$$\{0, 1\}$$

(b) What is the preimage of 1? You do not need to show your working.

(2 marks)

$$[1, \sqrt{2}) \cup (-\sqrt{2}, -1]$$

(c) Is f one-to-one? You should justify your answer with an explanation.

(3 marks)

$$\text{No: } f(0) = f(0.1) = 0$$

(d) Is f onto? You should justify your answer with an explanation.

(3 marks)

$$\text{No: } x^2 \geq 0 \quad \forall x,$$

$$\therefore \text{no } x \text{ has } f(x) = -1.$$

6. (a) Prove that the interval $[0, 1]$ has the same cardinality as the set $\mathbb{R}^+$ of all positive real numbers.

(Hint: You might wish to use the Schröder-Bernstein theorem.)

(5 marks)

1-1 func $f: [0, 1] \rightarrow \mathbb{R}^+$:

$$f(x) = x + 1.$$

$$\begin{aligned} \text{1-to-1 since } f(x) = f(y) &\rightarrow x+1 = y+1 \\ &\rightarrow x = y. \end{aligned}$$

1-1 func $g: \mathbb{R}^+ \rightarrow [0, 1]$

$$g(x) = \frac{1}{x+1}.$$

$$\begin{aligned} \text{Note: } x > 0 &\therefore g(x) > 0 \text{ and } g(x) < \frac{1}{0+1} = 1 \\ &\therefore g(x) \in [0, 1]. \end{aligned}$$

$$\text{1-to-1 since } g(x) = g(y)$$

$$\begin{aligned} &\rightarrow \frac{1}{x+1} = \frac{1}{y+1} \rightarrow x+1 = y+1 \\ &\rightarrow x = y. \end{aligned}$$

So: By Schröder-Bernstein,

$$|[0, 1]| = |\mathbb{R}^+|.$$

6. (b) Does $|\mathbb{Z}| = |\mathbb{Q}|$? You do not need to show your working.

(1 mark)

Yes.

- (c) Does $|\mathbb{Z}| = |\mathbb{Z} \times \mathbb{Z}|$? You do not need to show your working.

(1 mark)

Yes.

- (d) Does $|\mathbb{R}| = |\mathbb{Q} \times \mathbb{Q}|$? You do not need to show your working.

(1 mark)

No.

7. Define a relation ρ on $\mathbb{R}$ by $x\rho y$ if and only if $x \times y > 0$.

Determine whether ρ is reflexive, whether ρ is symmetric, whether ρ is transitive, and whether ρ is antisymmetric. For each property, explain why or why not.

(8 marks)

Reflexive: no since $0 \times 0 \not> 0$
 $\therefore 0 \not\rho 0$.

Symmetric: yes, since $xy > 0 \rightarrow yx > 0$.

Transitive: yes, since $xy > 0 \wedge yz > 0$

$$\rightarrow xy \cdot yz > 0$$

$$\rightarrow xz \cdot y^2 > 0$$

$$\rightarrow xz > 0 \text{ since } y^2 > 0$$

(and $y^2 \neq 0$,

since we know $xzy^2 > 0$)

Antisymmetric: no since

$$1 \cdot 2 > 0, 2 \cdot 1 > 0$$

$$\therefore 1 \rho 2 \text{ and } 2 \rho 1, \text{ but } 1 \neq 2$$

8. (a) What is the coefficient of x^2 in the expansion of $(3 - x)^6$? You should give your answer as a single integer, and show your working.

(2 marks)

$$(-1)^2 \cdot 3^4 \cdot \binom{6}{2} = 81 \cdot 15 \\ = 1215.$$

- (b) How many one-to-one functions are there from $\mathbb{Z}_3$ to $\mathbb{Z}_5$? You should give your answer as a single integer, and show your working.

(3 marks)

$$\begin{aligned} f(0): & 5 \text{ choices} \\ f(1): & 4 \text{ choices left} \\ f(2): & 3 \text{ choices left} \\ \therefore & 5 \cdot 4 \cdot 3 = \underline{60} \end{aligned}$$

8. (c) How many onto functions are there from $\mathbb{Z}_5$ to $\mathbb{Z}_3$? You should give your answer as a single integer, and show your working.

(5 marks)

All functions. $3^5 = 243$, since 3
choices of $f(x)$
for each x .

functions whose range $\neq 0$: $2^5 = 32$
 $\quad \quad \quad \quad \quad \neq 1$: $2^5 = 32$
 $\quad \quad \quad \quad \quad \neq 2$: $2^5 = 32$

functions whose range $\neq 0, 1$: $1^5 = 1$
 $\quad \quad \quad \quad \quad 0, 2$: 1
 $\quad \quad \quad \quad \quad 1, 2$: 1

$\therefore$ By inclusion-exclusion:

Functions whose range $\neq$ something:
 $32 + 32 + 32 - 1 - 1 - 1$
 $= 93.$

$\therefore$ Onto: $243 - 93 = \underline{150}.$

9. Let G denote the group $(\mathbb{Z}_2 \times \mathbb{Z}_2, +)$. Here we have $(a, b) + (c, d) = (a + c, b + d)$, where addition is done modulo 2.

Let H denote the group $(\mathbb{Z}_5 - \{0\}, \times)$. Here multiplication is done modulo 5.

Note that both G and H contain exactly four elements.

- (a) Write out the full Cayley tables for G and H .

(4 marks)

G :

	0,0	0,1	1,0	1,1
0,0	0,0	0,1	1,0	1,1
0,1	0,1	0,0	1,1	1,0
1,0	1,0	1,1	0,0	0,1
1,1	1,1	1,0	0,1	0,0

H :

	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

- (b) Are the groups G and H isomorphic? Explain why or why not.

(3 marks)

no. H has an elt of order 4
(2).
 G has all elts having order
2 or 1.

10. (a) A graph has 10 edges and 6 vertices. Five of the vertices have degree 3. What is the degree of the remaining vertex?

(3 marks)

Handshake theorem: $2e = \sum \text{degrees}$

$$\therefore 20 = 5 \cdot 3 + x$$

← unknown degree

$$\therefore x = 5$$

- (b) Consider the graph illustrated below. Does this graph have an Eulerian trail? You should justify your answer with an explanation.

No.

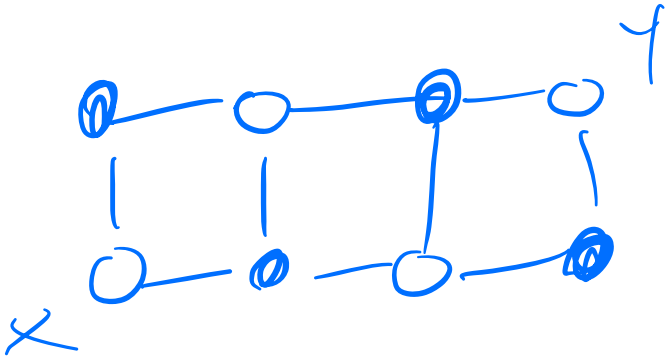
There are 4 vertices with odd degree.

(3 marks)

10. (c) Consider again the graph illustrated below. Prove that there is no walk of odd length that starts at vertex X and ends at vertex Y .

Colour vertices:

(4 marks)



Every step is black $\leftrightarrow$ white

$\therefore$ Odd walk from X must end on black.

But Y is white.

END OF EXAMINATION

MATH1061/MATH7861 Examination Formula Pages

Logical Equivalences

Given any statement variables p , q and r , the following logical equivalences hold, where **t** denotes a tautology and **c** denotes a contradiction:

<i>Commutative laws</i>	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
<i>Associative laws</i>	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
<i>Distributive laws</i>	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
<i>Identity laws</i>	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
<i>Negation laws</i>	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
<i>Double negative laws</i>	$\sim(\sim p) \equiv p$	
<i>Idempotent laws</i>	$p \wedge p \equiv p$	$p \vee p \equiv p$
<i>Universal bound laws</i>	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
<i>De Morgan's laws</i>	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
<i>Absorption laws</i>	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
<i>Negations of t and c</i>	$\sim \mathbf{t} \equiv \mathbf{c}$	$\sim \mathbf{c} \equiv \mathbf{t}$

Valid Argument Forms

Modus Ponens	$p \rightarrow q$ p $\therefore q$	Elimination	(a) $p \vee q$ $\sim q$ $\therefore p$	(b) $p \vee q$ $\sim p$ $\therefore q$
Modus Tollens	$p \rightarrow q$ $\sim q$ $\therefore \sim p$	Transitivity	$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$	
Generalization	(a) p $\therefore p \vee q$	Proof by division into cases	$p \vee q$ $p \rightarrow r$ $q \rightarrow r$ $\therefore r$	(b) q $\therefore p \vee q$
Specialization	(a) $p \wedge q$ $\therefore p$			(b) $p \wedge q$ $\therefore q$
Conjunction	p q $\therefore p \wedge q$	Contradiction Rule	$\sim p \rightarrow \mathbf{c}$ $\therefore p$	

Set Identities

Given any sets A , B and C that are subsets of a universal set U , the following laws hold:

- Commutative Laws (a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$
- Associative Laws (a) $(A \cup B) \cup C = A \cup (B \cup C)$
(b) $(A \cap B) \cap C = A \cap (B \cap C)$
- Distributive Laws (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- Identity Laws (a) $A \cup \emptyset = A$ (b) $A \cap U = A$
- Complement Laws (a) $A \cup A^c = U$ (b) $A \cap A^c = \emptyset$
- Double Complement Law $(A^c)^c = A$
- Idempotent Laws (a) $A \cup A = A$ (b) $A \cap A = A$
- Universal Bound Laws (a) $A \cup U = U$ (b) $A \cap \emptyset = \emptyset$
- De Morgan's Laws (a) $(A \cup B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = A^c \cup B^c$
- Absorption Laws (a) $A \cup (A \cap B) = A$ (b) $A \cap (A \cup B) = A$
- Complement of U and $\emptyset$ (a) $U^c = \emptyset$ (b) $\emptyset^c = U$
- Set Difference Law $A - B = A \cap B^c$

The Quotient-Remainder Theorem Given any integer n and any positive integer d , there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

Unique Factorisation Theorem Given any integer $n > 1$, there exists a positive integer k , distinct prime numbers $p_1, p_2, \dots, p_k$, and positive integers $e_1, e_2, \dots, e_k$ such that $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, and any other expression for n as a product of prime numbers is identical to this except perhaps for the order in which the factors are written.

Schröder-Bernstein Theorem For all sets A and B , if $|A| \leq |B|$ and $|A| \geq |B|$ then $|A| = |B|$.

Binomial Theorem Given any real numbers a and b , and any nonnegative integer n ,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$